A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 19

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** The dose that kills 50% of the (test) population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
01.2	**1 mark each** 1 mark for each of the 3 points below. **1 Insecticide X:** LD50 for the beetle (6) is very similar to that of the pest (5) so it has low specificity / kills the predator as well **1** Loss of the predator removes natural control so the pest population may recover faster / increase (resurgence) **1** Y has a much higher LD50 for the beetle (150) so few predators killed / more specific **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **LD50** → the dose that kills half of a test population **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** High water solubility so it is leached into rivers / groundwater / runoff **1** Toxic to aquatic organisms downstream **1** Mobile so it can affect areas away from the field **1** Washed off the crop quickly so more frequent application needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Long half-life = high persistence so it is not broken down quickly **1** So it remains in the organism / environment long enough to build up in tissues (especially if liposoluble) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	2
01.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Absorbed and translocated within the plant so protects new growth **1** Not washed off by rain **1** Does not need to cover all leaf surfaces / less spray needed **1** Less spray drift so less harm to non-target species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-264; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Σn(n - 1) = (40 x 39) + (30 x 29) + (20 x 19) + (10 x 9) = 1560 + 870 + 380 + 90 = 2900 **1** N(N - 1) = 100 x 99 = 9900 **1** D = 9900 / 2900 = 3.41 (accept 3.4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B **1** The index does not identify which species are present / a rare or important species may be missed – depends on identification skill and sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (2900 - 1400) / 1400 x 100 **1** = 107% (accept 107.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Large decrease from 24 per year (1975-79) to 2 per year (2010-14) **1** Steepest fall between the late 1970s and 1980s – slower decline / levelling off after 1995 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1** Double-hulled tankers required for all large tankers **1** Improved navigation (GPS, AIS) and offshore shipping routes **1** Inert gas systems reducing explosions **1** Twin engines / rudders reducing groundings after mechanical failure **1** Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) **1** Separate ballast tanks / crude oil washing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff **1 Routine operations:** small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) **1** Growth in oil use and transport means more small releases even if large accidents are rarer **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **diffuse** → pollution from a wide area with no single outlet	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1 Potable:** no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints **1 Spray irrigation:** low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food **1 Condenser cooling:** no gross solids - pipe blockage **1 Boiler water:** no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter **1** Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Clay particles carry electrostatic charges that make them repel each other so they stay suspended **1** Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Chlorine **1** Ozone **1** Ultraviolet light **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Molten iron from the blast furnace is converted directly to steel on the same site **1** So it does not cool and need re-melting – transport energy between sites is also avoided – waste heat from one process used by another (eg Kalundborg) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Collecting and processing small dispersed quantities can use more energy in transport than is saved **1** Mass production and large-scale mining have economies of scale – some materials (glass) take more energy to make than plastic but can be refilled many times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2
05.3	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared) – overhead cables cheaper than underground which need cooling **2** ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **pumped storage** → water pumped uphill when demand is low and released to generate at peak	4
05.4	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Proximity to the grid / demand to reduce transmission losses and cable costs **1** Access to cooling water for thermal stations **1** Resource availability (wind, tidal range, fuel deliveries) **1** Land availability and planning / public acceptance – stable geology **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	
06.1	**1 mark each** 1 mark for each of the 6 points below. **1 Direct effect of abiotic change:** species near the edge of their range of tolerance (temperature, water) may die out locally, eg sundews if marshes dry, frogs if climate drier – beech (shallow roots) suffers in drought more than oak **1 Changes in timing:** earlier flowering / emergence may separate flowers from pollinators – plant toxins building up earlier may kill caterpillars – warmer winters disturb dormouse hibernation so they use fat reserves and starve **1 Biotic changes:** food supply, predators, disease vectors (mosquitoes / malaria spreading), pollinators and seed dispersers may change **1 Distribution changes:** species may colonise newly suitable areas (triton molluscs around the UK) but slow-growing species (trees) may disappear from old ranges faster than they colonise – suitable areas or biological corridors may not exist – human land use blocks movement **1 Population fragmentation:** rising seas or upward retreat to mountains isolate populations – divided gene pools increase inbreeding – small populations may not be viable – no repopulation from elsewhere **1** Some species benefit (bats: warmer winters, more insects) but wetter stormier weather reduces feeding time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive • **Biotic** → living factors / effects of other organisms	6
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** All the species it depends on (food plants, pollinators, prey, mycorrhizae) will not move at the same speed **1** Less mobile species (plants, non-flying animals) cannot cross unsuitable areas without a biological corridor **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** The temperature difference between mid-latitude and polar air drives the jet streams – a smaller difference makes the jet stream slower and more meandering (Rossby waves) **1** Weather systems stall over an area, giving prolonged drought or rainfall / floods – unusually cold or warm air carried north or south **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 90-92; AQA spec 3.2.1.2.	
07.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Disposal of items containing mercury:** batteries, thermometers, fluorescent lights **1** Chemical plants producing chlorine using mercury electrodes **1** Combustion of coal **1** Gold mining (amalgamation) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
07.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses) **1** Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters **1** Disposal / storage of wastes at high pH to reduce solubility **1** Encapsulation of mercury-containing hazardous waste in cement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3

Do not write outside the box

Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are more soluble at low pH / less soluble at high pH **1** So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** The effects of two pollutants interact to produce a different (greater) effect than either alone **1** Cadmium and zinc – ozone and sulfur dioxide **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11.	
08.1	**1 mark each** 1 mark for each of the 2 points below. **1** In general, higher GDP per capita is associated with higher energy use per capita (positive correlation) **1 But not proportional:** B has a high GDP but much lower energy use than A or E – C uses relatively high energy for its GDP **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **correlation** → a relationship in which the two variables change together **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Affluent people buy more consumer goods, appliances and travel more / drive larger, less efficient vehicles and heat larger homes **1** Manufacture and transport of the extra products bought uses energy (indirect use) – larger houses use more energy to build **1** Affluent countries can afford energy for non-essential uses – cheap local energy (eg petrol in the USA) encourages wasteful use **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
08.3	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Higher energy prices / taxes encouraging conservation **1** Culture of energy conservation (eg Scandinavia after past shortages) / environmental awareness **1** Milder climate needing less heating or air conditioning **1 Service-based economy:** heavy industry (steel, aluminium) has moved to other countries so energy embodied in imports is not counted **1** Smaller / more efficient vehicles, better public transport, better insulated buildings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **energy conservation** → using less energy, eg by insulation or behaviour change **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Decline of heavy industries such as steel and aluminium smelting in the UK **1** The energy-intensive goods are now manufactured abroad (eg China) so the energy is counted in the exporting country **OR** efficiency improvements **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.5	1 mark for the point below. **1 Water:** pumping, treatment / sterilisation, sewage treatment – Agriculture: machinery, fertiliser manufacture (Haber process), irrigation pumps, processing / refrigeration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174; AQA spec 3.3.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Effluent gases pass through a chamber containing electrically charged wires / plates **1** Smoke particles become (negatively) charged and are attracted to the oppositely charged plates where they collect / stick **1** Plates are shaken and the particles fall as fly ash which is collected (and may be used, eg in building materials) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **electrostatic precipitator** → charged plates that attract and collect particulates	3
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Effluent gases are forced to rotate / spin in a cylindrical chamber **1** The (denser) particles are thrown outwards to the chamber walls where they fall and collect – cleaned gases leave through a central pipe **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.3	1 mark for the point below. **1** The tar / volatile organic compounds that produce smoke on incomplete combustion are driven off, producing smokeless coal / coke **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Turbochargers deliver more air / oxygen to the combustion chamber **1** So combustion is more complete and fewer unburnt carbon particles / soot are produced (especially during acceleration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Smaller particles remain suspended in the atmosphere for longer so they are more likely to be inhaled **1** They penetrate deeper into the lungs / alveoli and are not cleared – may carry toxic / carcinogenic chemicals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-247; AQA spec 3.4.3.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Uses / benefits:** nuclear electricity (low carbon, high energy density), medicine (X-rays, CT, radiotherapy, tracers, sterilising equipment), industry (thickness gauges, polymer cross-linking, testing turbines, oil exploration), agriculture (food irradiation, sterile male technique, mutation breeding), research tracers, smoke detectors, ship / submarine propulsion • **Risks:** free radical production, DNA damage, mutation, cancer – acute and chronic, somatic and gonadic effects – contamination vs exposure – activation products – long-lived wastes – accidents (Chernobyl, Fukushima) – weapons fallout • **Factors affecting risk:** type of radiation (alpha, beta, gamma, neutrons - penetration, RBE / weighting factors), half-life, dose (Gy, Sv), route of exposure • Risk:benefit analysis and why it is difficult (latency, alternative causes, poor exposure data, risks and benefits to different people, offsets such as avoided surgery) • **Principles:** ALARA, BATNEEC – closed sources, remote handling, absorbers / shielding, distance (inverse square law), time, protective clothing, decontamination, potassium iodide • **Monitoring:** dosemeters, film badges, air and contamination monitors – environmental monitoring via CPA and CGM – waste categories and disposal (vitrification, encapsulation, Drigg) • **Evaluation / judgement:** benefits generally outweigh risks when controls are applied – public perception vs actual risk (background radon 50% of dose vs <1% from nuclear industry)	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **half-life** → the time for the activity, or the amount, to fall by half • **vitrification** → sealing waste into glass **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	
	Model answer - Level 3 (8 marks)	
	Ionising radiation is used because it does things nothing else can: nuclear fission generates low-carbon electricity from fuel of extraordinary energy density; medicine relies on X-rays and CT scans, radiotherapy, tracers and gamma sterilisation of equipment; industry uses thickness gauges, cross-links polymers, tests turbine blades and explores for oil; and agriculture uses irradiation to preserve food and sterilise insect pests. The risks arise because radiation produces free radicals that damage DNA, causing mutation and cancer, with acute and chronic, somatic and gonadic effects; contamination is worse than exposure because the source stays with the person; and activation products, long-lived wastes, accidents at Chernobyl and Fukushima and weapons fallout add environmental risks.	
	Risk depends on the type of radiation - alpha is barely penetrating but has a weighting factor of 20 - on half-life, on dose in grays and sieverts and on the route of exposure. Weighing risks against benefits is difficult because cancers appear years later with many other possible causes, exposure data are poor, the people who benefit are often not those at risk, and benefits such as avoided surgery are hard to quantify.	
	Control follows ALARA and BATNEEC: closed sources, remote handling, shielding, distance under the inverse-square law, limiting time, protective clothing, decontamination and potassium iodide against iodine-131. Workers wear dosemeters and film badges and pass contamination monitors; discharges are managed by Critical Pathway Analysis and Critical Group Monitoring; wastes are vitrified, encapsulated or stored at licensed sites. With these controls the benefits generally outweigh the risks - natural radon supplies about half the average dose and the nuclear industry under one per cent - although public perception of risk remains far higher than the measured risk.	
	Weaker answer - Level 2 (5 marks)	
	Ionising radiation is used in nuclear power stations to make electricity, in hospitals for X-rays and treating cancer, and in industry. The risks are that radiation damages DNA and causes cancer, and accidents like Chernobyl release radiation into the environment. Nuclear waste stays dangerous for a long time. Workers are protected by shielding, keeping their distance and wearing film badges to measure their dose. The public is protected by monitoring around nuclear sites. Overall the benefits are worth the risks if it is controlled properly.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 2 covers uses, risks, controls and a judgement, but each briefly; Level 3 adds the factors that determine risk (type, half-life, dose units, route) and why risk-benefit analysis is difficult.	
	• Level 3 names the principles (ALARA, BATNEEC, inverse square law, CPA / CGM) rather than describing practice loosely.	
	• The judgement in Level 3 is supported with evidence (radon vs nuclear industry doses); Level 2's is an opinion.	
10.2	1 mark for the point below. **1** Consumers of the electricity may live in a different country / region from the uranium miners or the communities near waste stores who bear the risks **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 295-304; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration – hydrological cycle evaporation-precipitation • **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • **Tipping points:** self-sustaining processes – why control must precede them – examples • **Implications:** urgency and precautionary principle – need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation – geoengineering risks • Difficulty of prediction because feedbacks interact at different scales – models – international agreements and per capita footprints • Judgement on whether current strategies are adequate	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** energy density, renewability, abundance, location, intermittency / predictability, storage, transport, conversions, technology level, environmental impacts • **Fossil fuels:** choosing lower-impact fuels (gas over coal), desulfurisation, CCS, efficient combustion, siting power stations near cooling water and coal ports – new extraction technologies (directional drilling, secondary / tertiary recovery) and their impacts (fracking, tar sands) • **Nuclear:** high energy density reduces transport – waste properties (half-life) determine storage – thorium and fusion prospects • **Renewables:** matching resource to location (wind, HEP, solar, wave, tidal, geothermal, biofuel siting) – intermittency managed by storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management and diversity – low-impact designs (helical turbines, VAWT, tidal lagoons) • **Energy conservation reduces demand:** transport, buildings, industry (CHP, heat exchangers) • Impacts before and during use	25
	– life-cycle / embodied energy • **Evaluation:** no resource is impact-free – knowledge allows the best mix and mitigation – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

Do not write outside the box